

Searching for Changing-Look AGN Candidates in Repeat Optical Spectra

An optical spectroscopy project begun in summer 2025

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Abstract

A changing-look active galactic nucleus (AGN) can show a striking change in its optical continuum and broad emission-line spectrum between observations. During summer 2025, I explored a search for such changes using repeat Sloan Digital Sky Survey (SDSS) quasar spectra. This work included catalog selection, comparison of earlier and later observations, spectrum and difference-spectrum plots, and an exploratory area cut for visual screening. The project updates record the selection stages and contain several plots reproduced in this report. The aim was to identify spectra worth examining more closely for broad-line changes. The accompanying Python package develops this approach into a reproducible workflow for selecting pairs, reading FITS spectra and quality information, measuring emission lines, fitting simple broad Balmer components, and producing review plots. The package was reconstructed after the summer project. This report presents the summer exploration, the current methods, and the scientific checks needed before interpreting a review flag as a changing-look candidate.

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1 Question and physical background

The practical question is whether two optical spectra of the same AGN show a change that deserves closer examination as a possible changing-look event. An AGN spectrum combines a continuum associated with accretion, stellar light from the host galaxy, broad emission from gas with large line-of-sight velocities, narrow emission from more extended gas, and sometimes absorption or blended features. $H\beta$ lies near 4861 Å, $H\alpha$ near 6563 Å, and the usually narrow [O III] doublet near 4959 and 5007 Å. Broad and narrow components overlap, so a change in the total height of a line need not be a change in the broad-line region.

Broad-line appearance or disappearance is a useful optical signature of a changing-state AGN. Ordinary continuum variability, relative flux-calibration differences, changing amounts of host light inside the fiber, sky-subtraction residuals, and incomplete wavelength coverage can imitate aspects of a transition. Variable obscuration can also change the observed spectrum. Optical spectra alone may not decide between changing accretion state and changing obscuration; temporal, photometric, and multiwavelength information help [3, 2]. The code developed here therefore produces *review priorities*, not definitive physical classifications.

The project began with an interest in line identification and AGN spectral types. A 2025 presentation discussed FWHM, equivalent width, continuum fitting, and types 1, 1.5, 1.8, 1.9, and 2. Those slides show a direction being considered, not a completed general line finder or type classifier. The project-updates notes then shifted to repeat SDSS quasar spectra and direct comparisons. That narrower task is the scope of the later implementation.

2 Summer 2025 spectral exploration

2.1 Catalog selection and exploratory score

The notes describe the DR16Q quasar-only catalog [1], with a primary observation and additional plate–MJD–fiber identifiers for repeat spectra. The proposed sequence was to select $z < 0.8$, require at least one additional spectrum, discard invalid duplicate identifiers, and compare the observations with the smallest and largest MJD. Table 1 records the numbers written in the notes. These are the counts recorded during the summer exploration; they have not been independently reproduced with the current package.

Table 1: Selection and trial figures written in the 2025 project updates. These numbers have not been independently reproduced here.

Stage described in notes	Reported count
DR16Q catalog quasars	750,414
After the $z < 0.8$ cut	76,565
With at least one additional spectrum	8,141
Trial pairs processed before runtime became limiting	about 200
Trial plots above an area value of 1000	33

The early comparison subtracted the fluxes and investigated an area under the difference spectrum. The notes first discuss the whole range, then a rest-frame 4600–5000 Å interval near $H\beta$, with an exploratory cut of 1000. They describe taking an absolute value *after* integrating the signed difference. This is not generally the same as integrating the absolute difference:

$$A_{\text{signed}} = \left| \int_W [f_2(\lambda) - f_1(\lambda)] d\lambda \right|, \quad A_{\text{absolute}} = \int_W |f_2(\lambda) - f_1(\lambda)| d\lambda. \quad (1)$$

Positive and negative changes can cancel in the first quantity. Both have integrated-flux units if the flux-density and wavelength conventions are fixed. Neither has a universal threshold without a

specified window, grid, noise model, normalization, and empirical validation. The value 1000 was a trial plotting cut, not an established classifier.

2.2 Spectrum and difference-spectrum plots

The exported project-updates PDF contains plots from the original exploration. These plots document the spectrum visualization and two-epoch comparisons carried out during the summer work. The embedded images were extracted without redrawing the data, and their original labels are preserved. Figure 1 is described as redshift-adjusted in nearby prose, but its horizontal axis reads *Observed Wavelength*. Its frame cannot be settled from the image, so even apparently familiar line markers should not be treated as verified identifications.

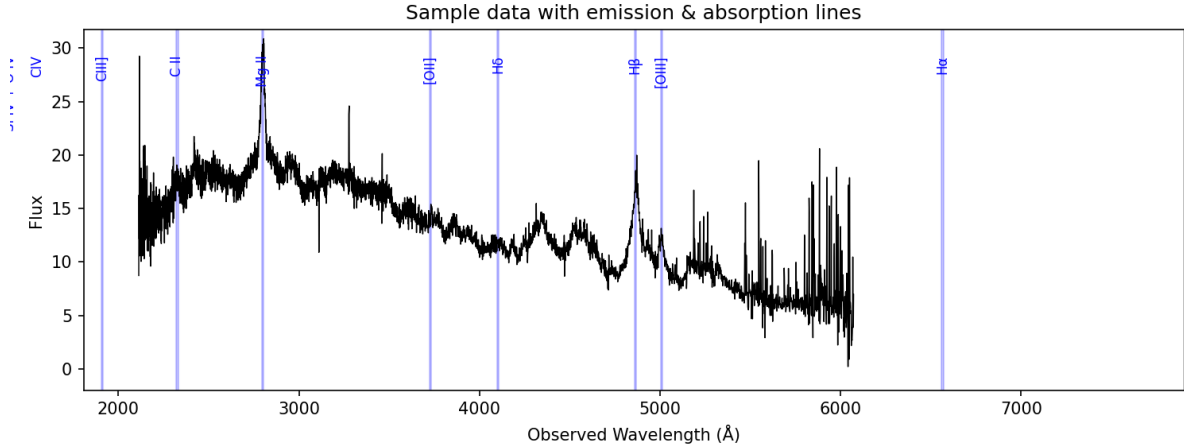


Figure 1: Spectrum plot embedded in the 2025 project updates. This is a historical output, not one generated by the current package. The conflict between the surrounding redshift-adjusted description and the *Observed Wavelength* axis cannot be resolved without the observation metadata. Source: author’s unpublished project-updates document, page 7.

The difference in Figure 2 is positive over much of the displayed interval. It illustrates why a large integrated difference can come from the continuum or calibration rather than a broad-line transition. Figure 3 has strong, narrow-looking residuals near marked features as well as a baseline difference. Its title puts a decimal value after *ObjID* that resembles a redshift, so the object identification shown is not independently reliable. Neither plot is classified as a changing-look AGN here.

The notes mention two NumPy arrays of earliest and latest identifiers and a later slowdown in downloads or processing. The current reproducibility record is described in the repository documentation. Papers on double-peaked [O III] binary-AGN *candidates* and recoiling black holes were background reading, not completed analyses here. The earlier spectral-stacking project is separate and is not used to claim results in this report.

3 Repeat-spectrum analysis methods

3.1 Inputs and repeat-observation manifest

The new package expects the SDSS DR16Q_v4.fits quasar-only catalog, documented by SDSS [1, 4]. The `select-pairs` command reads *Z*, *NSPEC*, the primary plate/MJD/fiber, and the duplicate arrays. It requires finite $0 < z < 0.8$ by default and $\text{NSPEC} \geq 1$, keeps only positive observation identifiers, removes exact duplicate triples, and chooses the earliest and latest distinct MJDs. It writes one CSV row per selected catalog object. The rest-frame separation is

$$\Delta t_{\text{rest}} = \frac{\text{MJD}_{\text{late}} - \text{MJD}_{\text{early}}}{1 + z}. \quad (2)$$

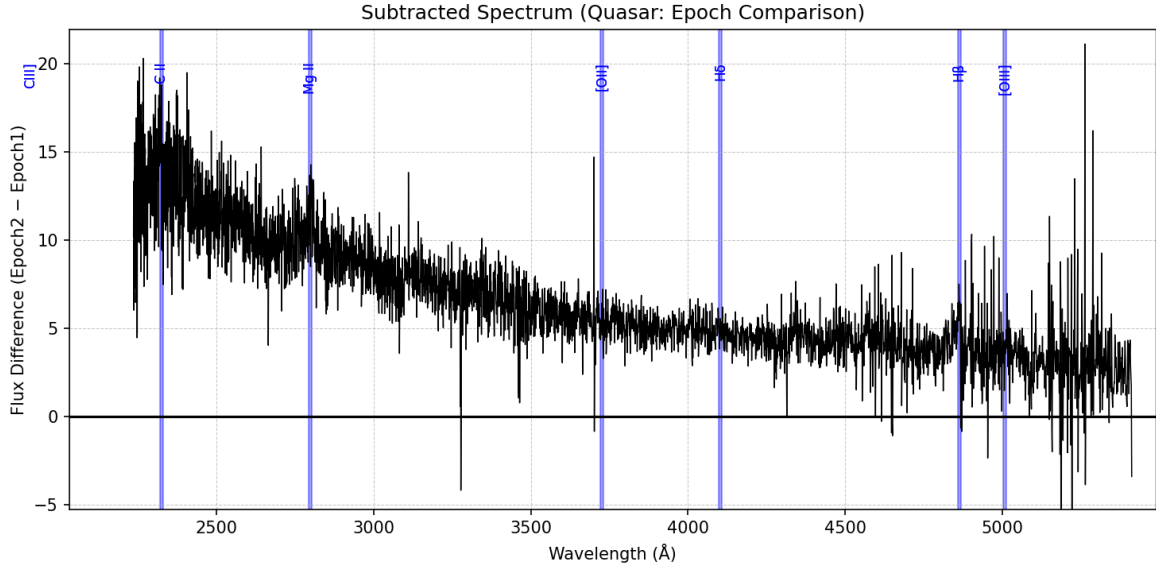


Figure 2: Early two-epoch flux-difference plot from the project updates. A broad continuum offset is visible; object identity, calibration, pixel masks, and error arrays cannot be recovered from the screenshot. Source: author's unpublished project-updates document, page 8.

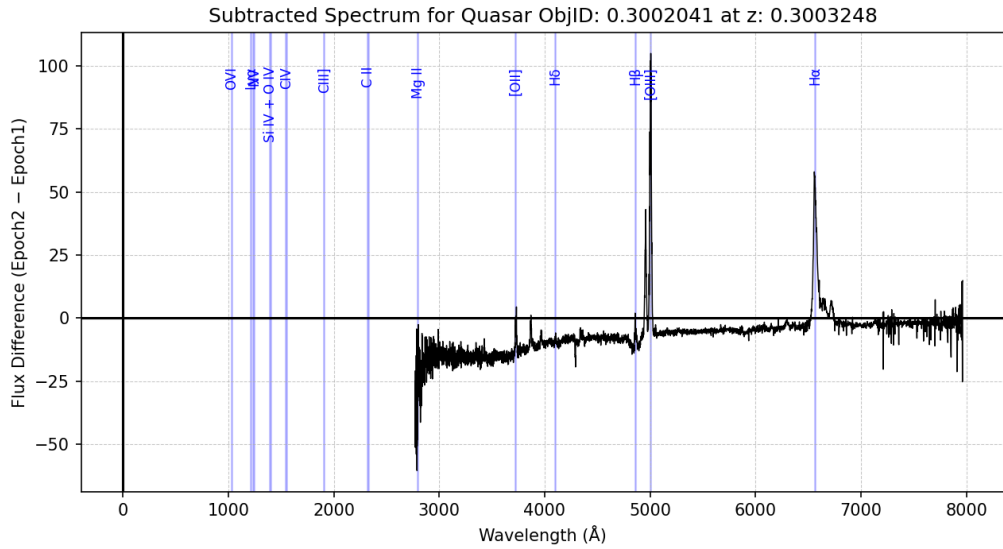


Figure 3: Another historical difference plot, selected from the sample images in the notes. Residual spikes near marked features invite inspection, but the original data and the plot's object identifier are not verifiable. It is not evidence of a confirmed broad-line transition. Source: author's unpublished project-updates document, page 13.

This gives context for variability, but two epochs do not reveal when a transition occurred. Selecting a quasar-only catalog can miss objects classified differently in another epoch, so the input itself limits completeness.

The redshift cut helps retain $H\beta$ in optical coverage, but does not guarantee $H\alpha$. At $z = 0.8$, $H\beta$ appears near 8750 Å, while $H\alpha$ would lie near 11800 Å. Actual usable coverage depends on the instrument and data quality; the code checks each line separately. The notes’ rough safe redshift limits were working estimates, not universal spectrograph boundaries.

The optional `fetch` command uses `astroquery` to retrieve individual SDSS spectra by plate, MJD, fiber, and release [5]. A local cache avoids repeat retrieval. A log records the requested release, retrieval time in UTC, SHA-256 file hash, and errors. Survey spectra are not redistributed in the repository. A failed download remains a logged failure. The live SDSS retrieval path has not been demonstrated here on a full real-data sample.

3.2 Wavelengths, fluxes, and usable pixels

For a standard individual SDSS spectrum, the reader uses HDU 1 columns `loglam`, `flux`, and `ivar`. It forms

$$\lambda_{\text{obs}} = 10^{\text{loglam}} \text{ Å}, \quad \lambda_{\text{rest}} = \frac{\lambda_{\text{obs}}}{1+z}. \quad (3)$$

The stored flux density remains in its observed-frame SDSS scale. Its conventional unit is

$$10^{-17} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ Å}^{-1}.$$

Dividing wavelength by $1+z$ does not turn the flux density into a rest-frame luminosity density. Rest-frame coordinates determine the line windows; observed-wavelength widths determine observed integrated line flux.

The reader rejects nonfinite flux, nonfinite or nonpositive inverse variance, and pixels with a nonzero `and_mask` when that column is present. Selected `or_mask` bits may also be rejected explicitly, but the default does not reject every OR bit. Bad pixels and gaps are especially important near a narrow line or sky residual. The implementation requires valid sidebands and line windows without large gaps. This is a starting rule; a real-candidate review should examine the SDSS reduction, wavelength solution, spectral resolution, sky residuals, and flux calibration.

3.3 Continuum and integrated line flux

A local continuum is fitted separately in each epoch for each line. If $x_i = \lambda_{\text{rest},i} - \lambda_0$ and f_i is the observed-frame flux density, the model in two sidebands is

$$C_i = a + bx_i. \quad (4)$$

The coefficients minimize $\sum_{i \in C} \text{ivar}_i (f_i - a - bx_i)^2$ over valid sideband pixels C . With design matrix X and diagonal inverse-variance matrix W , the formal coefficient covariance is $(X^T W X)^{-1}$. The implementation scales it upward when the sideband reduced χ^2 exceeds one. This accounts approximately for scatter in the sidebands, but not correlated pixels, curvature, or calibration systematics.

For line pixels in window L , trapezoidal weights u_i are calculated on that spectrum’s own rest-wavelength samples. Since $d\lambda_{\text{obs}} = (1+z)d\lambda_{\text{rest}}$, observed line flux is

$$F_{\text{line}} = \sum_{i \in L} (1+z) u_i [f_i - (a + bx_i)]. \quad (5)$$

In standard SDSS units the numerical result is in $10^{-17} \text{ erg s}^{-1} \text{ cm}^{-2}$. Let $v_i = (1+z)u_i$ and $\mathbf{q} = (\sum_i v_i, \sum_i v_i x_i)^T$. The formal variance implemented is

$$\sigma_F^2 = \sum_{i \in L} \frac{v_i^2}{\text{ivar}_i} + \mathbf{q}^T \text{Cov}(a, b) \mathbf{q}. \quad (6)$$

The terms represent line-pixel noise and fitted-continuum uncertainty, assuming independent pixels. For a positive fitted continuum at line center, rest equivalent width is

$$\text{EW}_{\text{rest}} = \frac{F_{\text{line}}}{(1+z)a}. \quad (7)$$

This is not a pure broad-component EW when narrow or blended features lie inside the window.

Table 2: Rest-frame windows in Å used by the package. H β , H α , and Mg II follow Guo et al. (2024), Table 3 [2]; the [O III] 5007 diagnostic window is a local implementation choice.

Feature	Line	Left sideband	Right sideband
H β	4780–4940	4730–4770	5030–5080
H α	6450–6700	6420–6450	6750–6790
Mg II	2750–2850	2680–2720	2880–2920
[O III] 5007	4996–5018	4970–4990	5025–5045

H α ’s broad integration window contains [N II] and narrow H α ; Mg II can blend with Fe II. The [O III] 5007 measurement is only a warning diagnostic: a sufficiently significant and substantial change may suggest calibration or aperture trouble, although genuine narrow-line variability is possible. It is not an automatic normalization anchor.

3.4 Broad Balmer fits and review flag

For H β and H α , the code compares a narrow-only model with a narrow-plus-broad model. Each has a linear continuum and a narrow Gaussian; the H α complex adds [N II] at 6548 and 6583 Å with a tied width and shift and a fixed 6583-to-6548 amplitude ratio of 2.96. A positive broad Gaussian near the Balmer center is

$$g_b(\lambda) = A_b \exp \left[-\frac{(\lambda - \lambda_b)^2}{2\sigma_b^2} \right], \quad \text{FWHM}_v = \frac{2\sqrt{2 \ln 2} \sigma_b}{\lambda_0} c. \quad (8)$$

Here σ_b and λ_0 are in rest Å and $c = 299792.458 \text{ km s}^{-1}$. The reported broad flux integrates the fitted Gaussian with a $(1+z)$ factor, because its amplitude remains an observed-frame flux density.

The fits use inverse-variance weighted least squares. With n fitted pixels and k parameters the code calculates $\text{BIC} = \chi^2 + k \ln n$ under an independent-Gaussian-error approximation. It provisionally detects a broad component when the fits converge, the broad model improves BIC by more than 10, formal broad-flux S/N is at least 5, FWHM is at least 1000 km s^{-1} , and the width is below its ceiling. A pair is flagged for *review* if provisional detection status changes and the fitted broad-flux change exceeds three combined formal standard deviations. An integrated-window change contributes to the separate priority score but does not alone set this flag.

These are heuristic cuts, not calibrated probabilities. The model omits host templates, Fe II, absorption, asymmetric profiles, detailed instrumental resolution, and epoch-to-epoch covariance. Calibration or differing host fractions can bias both continuum and broad fits. BIC and parameter covariance describe the adopted model, not the probability that a source is a changing-look AGN.

3.5 Grid alignment and diagnostic plots

The 2025 notes observed one pair whose wavelength arrays appeared to coincide and considered direct subtraction. This cannot be assumed for all pairs. The new measurements and fits use each epoch’s native grid, so numerical ranking does not subtract unmatched pixels. Only the diagnostic plot interpolates both valid spectra onto a common overlapping rest-wavelength grid. It leaves gaps blank when neighboring source pixels are invalid. The plotted difference has no propagated covariance and is never used as a detection statistic.

4 Checks and example output

The tests use constructed spectra where the answer is known. A Gaussian line with a flat continuum has analytic integrated flux $A\sigma\sqrt{2\pi}$; the implementation recovers it to the test’s numerical tolerance at $z = 0$ and $z = 0.6$ while preserving rest EW. A continuum-only offset is designed not to trigger a line-transition flag. Other checks cover different epoch grids, missing $H\alpha$ coverage, masked line pixels, zero inverse variance, earliest/latest catalog selection, and a FITS round-trip. Eight tests passed locally and in the repository’s automated workflow for this version. These are implementation checks under simplified conditions, not measures of survey completeness.

The deterministic synthetic example has $z = 0.15$ and slightly offset grids. Its earlier spectrum contains injected broad $H\beta$ and $H\alpha$ Gaussians; the later one retains continuum and narrow features but not the broad components. Gaussian noise is added. Figure 4 shows both epochs and a display-only difference. The code flags the pair for review; integrated-window changes are approximately -27.1 and -24.1 *formal* standard deviations for $H\beta$ and $H\alpha$. Their size follows from a deliberately clear injected signal and the independent-noise model. This is not an SDSS detection or an estimate of false-positive rate.

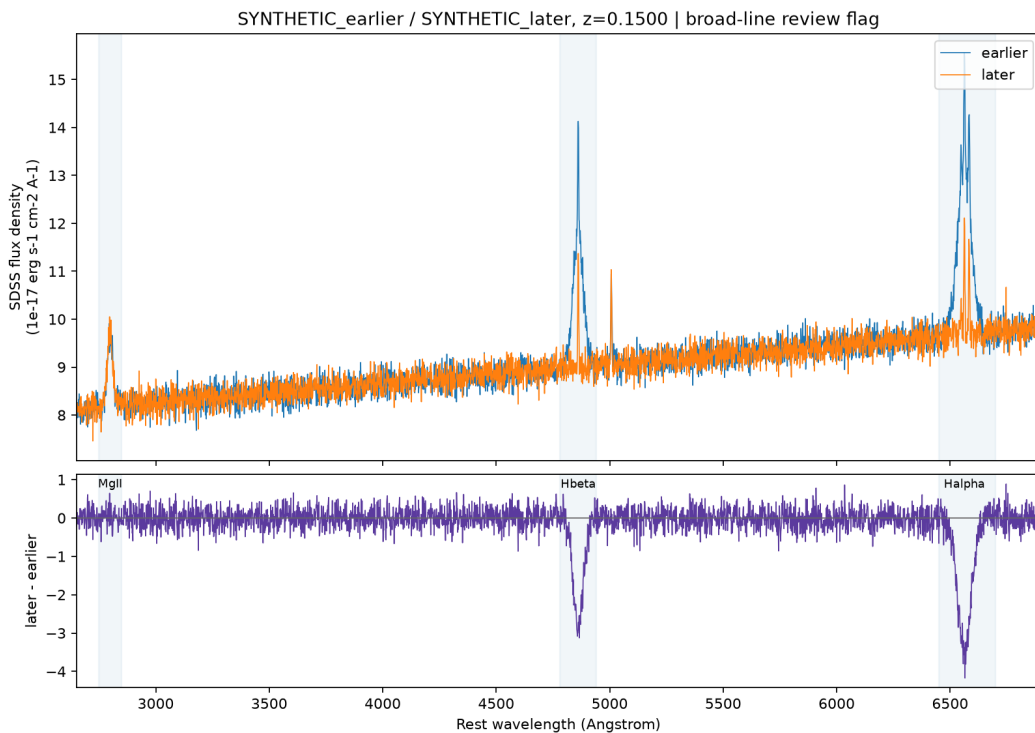


Figure 4: Synthetic example generated by the accompanying package. Blue is earlier, orange later, and purple is the interpolated display difference (later minus earlier). Broad Balmer emission was deliberately removed in the later epoch. These are not observed SDSS spectra.

5 Interpretation, limitations, and next steps

An interesting plot is only the beginning of candidate assessment. The original FITS files should be checked for adequate coverage and signal-to-noise in the continuum and line wings, redshift consistency, masks, sky residuals, resolution, and calibration. Both epochs should be examined separately with continuum and line models appropriate to the source. Narrow $[OIII]$ behavior may inform a relative-calibration check but is not an infallible standard. Further spectra and photometry can show whether the change persists or tracks the continuum. Multiwavelength data help assess obscuration and accretion interpretations [3, 2]. A convincing broad-line transition can be proposed as a candidate after such review; a confirmed classification needs stronger evidence

than a score crossing.

For this package to become a validated search, the next step is a controlled real-data run with catalog checksum and SDSS release recorded. Known changing-look examples and stable controls should be assembled independently. Visual review should sample low as well as high scores, not only apparent successes. Recovery and false-positive rates should be measured versus signal-to-noise, redshift, line coverage, flux state, and rest-frame time baseline. Thresholds should be tuned on one sample and evaluated on another. The historical area cut of 1000 must not be used as a validated boundary. No real-object candidate list, exhaustive line detection, classification completeness, or physical mechanism is established here.

6 Conclusion

During summer 2025, I investigated repeat SDSS quasar spectra as a way to search for possible changing-look AGNs. The work progressed from studying spectral types to selecting repeat observations, comparing earlier and later spectra, producing difference plots, and exploring numerical screening followed by visual inspection. The catalog-selection record and spectrum figures presented here document that exploratory analysis. A central lesson was that a large flux-difference integral can reflect continuum variability or calibration as well as emission-line changes, so line-specific measurements and careful inspection are essential.

The accompanying package provides a reproducible way to develop this approach through quality checks, local continuum measurements, simplified broad Balmer fits, and diagnostic plots. Its controlled examples check the numerical behavior of the implementation. Applying these tools to a reviewed sample of real spectra and calibrating the selection criteria are the next steps toward a validated candidate search. The project provides an exploratory analysis and a practical basis for further study; it does not claim confirmed changing-look discoveries.

Reproducibility and source

The Python package and compiled report are at the Changing-Look AGN Detection repository on GitHub. The historical figures were extracted from the author’s unpublished 2025 project updates and included as examples of the summer analysis. Survey catalog and spectra should be retrieved from SDSS rather than redistributed here.

References

- [1] B. W. Lyke et al., *The Sloan Digital Sky Survey Quasar Catalog: Sixteenth Data Release*, The Astrophysical Journal Supplement Series **250**, 8 (2020). doi:10.3847/1538-4365/aba623.
- [2] W.-J. Guo et al., *Changing-look Active Galactic Nuclei from the Dark Energy Spectroscopic Instrument. I. Sample from the Early Data*, The Astrophysical Journal Supplement Series **270**, 26 (2024). doi:10.3847/1538-4365/ad118a.
- [3] C. Ricci and B. Trakhtenbrot, *Changing-look active galactic nuclei*, Nature Astronomy **7**, 1282–1294 (2023). doi:10.1038/s41550-023-02108-4.
- [4] Sloan Digital Sky Survey, DR16Q_v4 Data Model, https://data.sdss.org/datamodel/files/BOSS_QSO/DR16Q/DR16Q_v4.html.
- [5] Astroquery developers, SDSS Queries (astroquery.sdss), <https://astroquery.readthedocs.io/en/latest/sdss/sdss.html>.